[OL] 0 Abstract

This paper investigates the role of self-reflection in reinforcement learning by modeling reflection as an explicit action for a tabular Q-learning agent. When the agent selects the reflect action, the environment state remains unchanged, a reflection cost is subtracted from the reward at that step, and the agent's exploration and learning rates are updated based on recent performance. This approach incorporates the cost of reflection directly into the Markov Decision Process, facilitating controlled comparisons between a standard Q-learning baseline and reflective agents subjected to varying reflection costs. A 20×20 GridWorld navigation task is used to evaluate episodic return and reflection frequency, thereby identifying conditions under which reflection enhances learning efficiency and examining how reflection cost affects the agent's willingness to reflect.

This paper examines the role of self-reflection in reinforcement learning by modeling reflection as an action for a tabular Q-learning agent. Selecting the reflect action leaves the environment state unchanged, subtracts a reflection cost from the reward at that step, and updates the agent's exploration and learning rates based on recent performance. This formulation adds the cost of reflection directly into the Markov Decision Process and enables controlled comparison between a standard Q-learning baseline and reflective agents operating under different reflection costs. Using a 20×20 GridWorld navigation task, we evaluate episodic return and reflection frequency to determine when reflection improves learning efficiency and how its cost influences the agent's willingness to reflect.

[OL] 1 Introduction

edited verison, removing some details

Reinforcement learning (RL) studies how an agent can learn to behave optimally through trial and error by maximizing expected cumulative reward in an environment. In standard RL, the agent interacts with the environment over a sequence of states, selects actions, receives rewards, and updates its policy accordingly.

This project investigates the role of self-reflection in a controlled RL setting. In our formulation, reflection is modeled as an explicit action available to the agent. When the agent executes the reflect action, the environment state does not change; instead, the agent performs an internal update that modifies its learning parameters and incurs an immediate cost, which is subtracted from the reward. This penalty represents the cost of allocating a decision step to introspection rather than to environmental interaction, creating a direct trade-off between immediate and potential long-term benefit.

Our goal is to determine whether introducing this reflection action leads to a net improvement in learning outcomes relative to a standard tabular Q-learning baseline. To isolate the impact of reflection, we compare two agents that are identical except for the availability and use of the reflect action. The baseline agent applies tabular Q-learning with an ϵ\epsilonϵ-greedy exploration strategy and fixed learning parameters, whereas the reflective agent may choose to reflect; upon reflecting, it adjusts internal parameters based on recent performance.

We evaluate the reflective agent under varying reflection costs in a GridWorld environment, measuring episodic return and reflection frequency. These metrics directly address two key questions: under what conditions does self-reflection improve performance despite its cost, and how does the cost of reflection influence the agent's decision to reflect. We hypothesize that in environments where standard directional actions are sufficient to learn an effective policy, reflection may impose an unnecessary penalty. In contrast, in settings where learning is unstable or requires adjustment, reflection may improve performance by guiding exploration and learning-rate adaptation.

Reinforcement learning (RL) studies how an agent can learn to behave optimally through trial and error by maximizing expected cumulative reward in an environment. In the standard formulation, the agent interacts with the environment over a sequence of states, selects actions, receives rewards, and updates its policy to improve future decisions. While this interaction typically focuses on choosing "external" actions that change the environment, many intelligent systems also engage in "internal" processes—such as reviewing recent performance or adjusting learning strategy—that consume resources but may improve long-term behavior.

This project investigates the role of self-reflection in a controlled RL setting. Because the term "self-reflection" can be interpreted in several ways, we adopt a precise, operational definition aligned with our implementation. When the agent chooses the reflect action, the environment state does not change; instead, the agent performs an internal update that modifies its learning parameters. Importantly, reflection is not assumed to be free: we model reflection as incurring an immediate scalar penalty, implemented by subtracting a cost c≥0c \ge 0c≥0 from the reward on that timestep. This cost represents the computational/time overhead of reflecting and creates a direct trade-off: reflecting may improve future learning behavior, but it can also reduce immediate return and waste decision steps if used unnecessarily.

Our goal is to determine whether introducing this explicit reflection action leads to a net improvement in learning outcomes relative to a standard tabular Q-learning baseline. To isolate the impact of reflection, we compare two agents that are identical except for the availability and use of the reflect action. The baseline agent applies tabular Q-learning with an ϵ\epsilonϵ-greedy exploration strategy and fixed learning parameters. The reflective agent uses the same Q-learning update rule but may choose to reflect; upon reflecting, it adjusts internal parameters—specifically the exploration rate ϵ\epsilonϵ and learning rate α\alphaα—based on recent performance. If recent rewards improve, the agent reduces exploration (lower epsilon-ϵ) and increases the learning rate (higher alpha-α); if rewards drop, it increases exploration (higher epsilon-ϵ) and decreases the learning rate (lower alpha-α) to avoid overreacting to noisy outcomes.

We evaluate the reflective agent under multiple reflection costs in a GridWorld pathfinding environment. We measure episodic return and reflection frequency. These metrics directly address two key questions: under what conditions does self-reflection improve performance despite its cost, and how does the reflection cost influence the agent’s decision to reflect. We hypothesize that in environments where standard directional actions are sufficient to learn an effective policy, reflection may impose an unnecessary penalty. In contrast, in settings where learning is unstable or requires adjustment, reflection may improve performance by guiding exploration and learning-rate adaptation.

Alternate Introduction

Self-reflection is an agent’s ability to inspect and summarize its own policy and experience. It is a technique that may enable faster adaptation to environments and better generalization by evaluating its own internal decision making process like policy, values estimates or recent trajectory. The purpose of the project is to determine if reflecting on past behavior is helpful and/or to what extent it is helpful when counter-weighted by a cost. When comparing agents with and without self-reflection, self-reflective agents consume more resources and propagate mistakes or biases depending on the articulation. Great advances have been made in AI technologies for gaining insight and understanding (Lewis and Sarkadi, 2024). However, limited research exists on employing self-reflection in improving problem-solving and learning. The primary aim of this research is to investigate the role of self-reflection, particularly on its utility in a single-agent AI system. We will present and evaluate a method for this from gridworld problems.

[OL] 2 Problem Formulation

2 Problem Formulation

We model the agent’s interaction with the environment as a finite and episodic Markov Decision Process (MDP). We chose this model because it is the standard framework for formalizing problems in reinforcement learning. We also include a discount factor γ to modulate the agent’s preference for short-term rewards over long-term rewards; these models are sometimes referred to as Discounted Reward Markov Decision Processes.

Formally, our model is a 5-tuple (S, A, P, R, γ). Here, S denotes the finite state space and A denotes the set of available actions. P is a probability function that defines the distribution of states for a given action (transition dynamics); we have $INSERT P IN MATH HERE$. The reward function R is defined as $INSERT R STUFF HERE$. Finally, $INSERT GAMMA IN 0-1$ the discount factor.

The agent seeks to learn a policy $INSTERT POLICY IN MATH$ that maximizes the expected return over an episode. To incorporate self-reflection into this MDP, we include a reflection action. So, for a gridworld task, the action set will be $MAKE MATH A={up, down, left, right, reflect}$. When the agent chooses the reflect action, the state of the environment remains unchanged $INSERT P STATE ACTION STUFF$. However, reflection incurs an immediate penalty that reduces the reward received on that timestep. This cost is implemented as $INSERT r <-r-c$ where $c \ge 0$ is a scalar reflection cost.

Let $G_reflect$ denote the episodic return of a reflective agent and $G_baseline$ the return of the baseline agent, without the ability to reflect. We want to determine under what conditions,

$INSERT EXPECTED VALUE THING$

And how the reflection cost c influences this relationship. We hypothesize that when the transition and reward structure allows the agent to improve its return through selecting direction actions alone, reflection may impose an unnecessary cost. In contrast, environments that demand greater adaptation may benefit from reflection.

We evaluate performance using three metrics:

1. Episodic Returns, which is the returns per episode of each agent, and measures the performance by comparing path completion progress against cost-incurring actions.
2. Reflection Frequency, which is the reflection usage throughout the episodes of each agent, and shows how reflection cost influences the agent’s decisions and resulting returns.
3. Success Rate, which measures performance as the proportion of episodes that the agent reaches the goal position.

[OL] 3 Related Work

3. Related Work

In general, we found that definitions of reflection and self-reflection were not standardized. As such, much of the related work was consulted to learn about different formulations of reflection and how these formulations were evaluated. Much of this work seeks to solve problems that are not directly related to ours. In particular, a large part of the related work involved studying various notions of reflection in large language models.

Cox (2005) broadly surveys the field of metacognition and proposes various definitions for reflection. However, they do not formalize the cost of reflection as it relates to selecting actions. Reflection is also assumed to always be beneficial to cognitive systems; this assumption is not tested empirically.

Omidshafiei et al. (2018) allow agents to learn when and what advice to give to peers to accelerate group learning. Under their formulation, a teacher agent is allowed to send advice signals to student agents in a multi-agent reinforcement-learning setting. The study shows that judicious teaching leads to a significant speedup in cooperative learning. However, the formulation treats advice as an arbitrary signal rather than a self-articulation of an agent's own policy. A cost for giving or receiving advice is not modelled. The formulation also assumes advice is generally beneficial and does not analyze the possible performance decrease from receiving badly formulated or misguided advice.

Renze and Guven (2024) examine the impact of learning from past traces on the problem-solving performance of large language model agents. By comparing baseline prompts with reflection-enabled prompts across a range of reasoning tasks, they show that reflection often increases accuracy and helps detect mistakes in single-episode reasoning. However, this work is limited to online or single-agent language tasks and does not address sequential decision making with explicit episodic returns. In addition, they do not explicitly model a cost for producing reflections such as computational, time, or energy penalty. Finally, they do not explore how self-reflection may influence other agents or propagate errors in a multi-agent setting.

Zhang et al., (2024) propose using reflection as a mechanism to adjust the internal policy components of LLMs to improve performance after failure. Under their formulation, an LLM agent articulates beliefs about the world and itself in natural language based on external performance feedback and its previous beliefs. These are included as part of the subsequent prompt. In this way, the articulations act like a natural language extension of the agent's policy. Thus, the agent considers aspects of its own policy as part of reflection. However, their model does not assign a cost to reflection, nor does it explore when reflection may degrade the performance.

Taken together, these lines of research show that various forms of self-reflection or advice can improve performance. However, most research in this area does not address the cost-benefit trade-offs and fidelity of choosing to self-reflect.

[OL] 4 Proposed Approach

4 Proposed Approach

Our aim is to determine whether adding an explicit reflection action to a standard reinforcement-learning agent improves learning efficiency despite its immediate cost. Following the problem formulation in Section 2, we structure the experiments to isolate reflection cost as the only differing component between agents, and use a clearly defined baseline for comparison. We extend a tabular Q-learning agent with a reflect action, that leaves the environment state unchanged, but modifies the agent's internal learning parameters while incurring a fixed penalty. This formulation allows us to evaluate the effect of reflection on learning performance, and if the benefit produced by reflection outweighs its immediate cost.

The objective is to determine whether incorporating an explicit reflection action into a standard reinforcement-learning agent enhances learning efficiency despite its immediate cost. Consistent with the problem formulation in Section 2, the experiments are structured to ensure that reflection cost is the sole variable distinguishing agents, with a clearly defined baseline for comparison. A tabular Q-learning agent is extended with a reflect action, which leaves the environment state unchanged but modifies the agent's internal learning parameters while incurring a fixed penalty. This approach facilitates assessment of the impact of reflection on learning performance and evaluation of whether the benefits of reflection outweigh its immediate costs.

4.1 Baseline Agent

The baseline agent is a standard tabular Q-learning agent with an e-greedy exploration strategy. It receives no other guidance or heuristics. The baseline agent updates its action-value estimates using the Q-learning rule

(INSERT QLEARNING UPDATE FORMULA)

Where alpha is the learning rate initialized as 0.3, gamma is the discount factor initialized as 0.95, and r is the reward obtained after executing action a in state s.

4.2 Reflective Agent

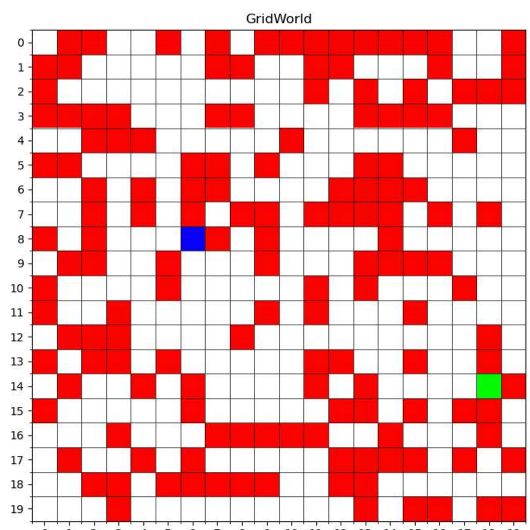
The reflective agent is identical to the baseline in its use of tabular Q-learning, with the exception that it may choose an additional reflect action. Executing the reflect action leaves the environment unchanged but imposed an immediate penalty on the reward, implemented as

(INSERT REFLECTION COST FORMULA)

Where c >= 0 and denotes the reflection cost defined in the MDP. We evaluate three cost levels, where c = {0.1,0.2,0.3}, to examine whether the cost of the reflection action has an influence on the agent's decision to reflect. Upon reflecting, the agent updates its internal learning parameters, specifically the exploration rate epsilon and the learning rate alpha. Specifically, the agent compares the average reward over its most recent five actions with the average reward from the preceding five. If the performance has improved, epsilon is decreased (BY HOW MUCH) and alpha is increased (BY HOW MUCH), therefore promoting more exploitative behaviour and placing greater weight on the new information. If performance has deteriorated, epsilon is increased and alpha is decreased, allowing for more exploration while reducing the influence of potentially misleading rewards.

4.2 GridWorld Environment

The experiments are conducted in a 20x20 GridWorld environment in which start and goal positions, as well as obstacle placements, are randomly placed to produce different maps. To generate variation in navigational complexity, we construct maps with three structural configurations: 100 obstacles with a minimum start–goal distance of 10, 150 obstacles with a minimum distance of 15, and 200 obstacles with a minimum distance of 20. We classify the maps into different degrees of difficulty based on baseline agent performance. For each map configuration, the baseline agent is trained for three independent runs of 300 episodes, and the resulting success rates are averaged to obtain a performance measure x. Maps for which the baseline achieves an average success rate greater than 95%, $x \ge 95%$ are labeled easy, those with$25%\le x < 95%$ are labeled medium, and those with $x <25%$are designated hard.



4.3 Experiment

We train three reflective agents, each with differing reflection costs, and the baseline agent on identical sequences of GridWorld environments. All agents interact with the same transition dynamics: movement actions advance the agent deterministically unless blocked by an obstacle, each step incurs a reward of −0.1, and reaching the goal yields a terminal reward of +10. Each agent is evaluated over 300 episodes, with a maximum of 1000 steps per episode. Throughout training, we record episodic return achieved by each agent as well as the number of times the agent chooses the reflect action. These metrics are collected across all map difficulty levels.

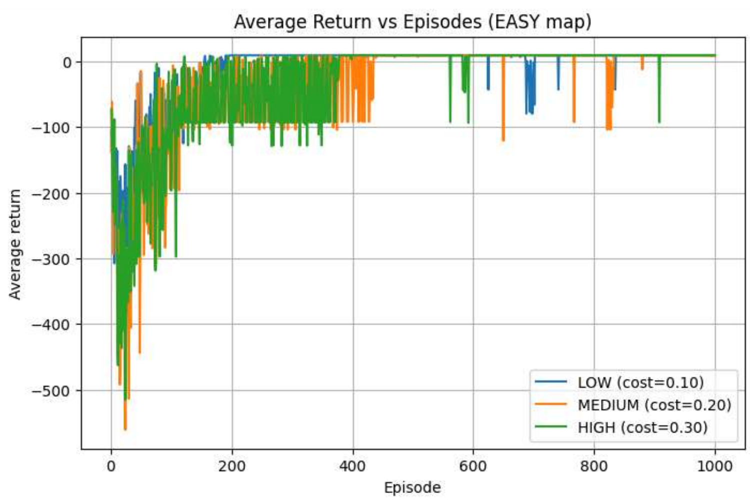
[OL] 5 Evaluation

5. Evaluation

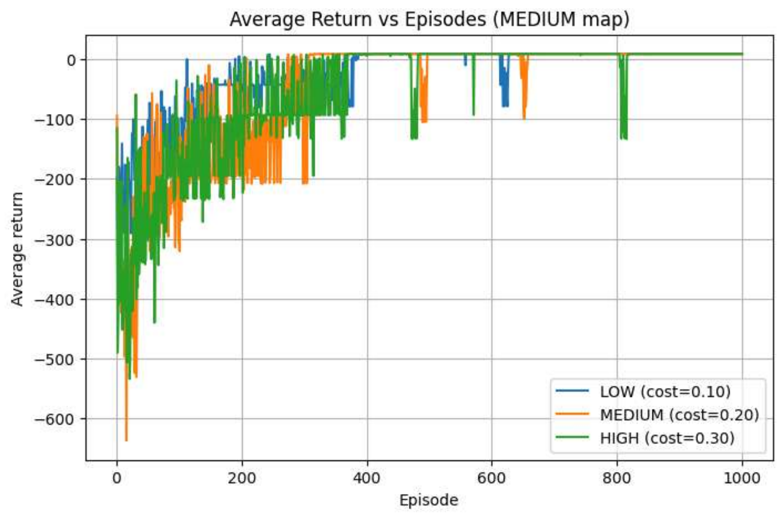
To evaluate the utility of self-reflection in reinforcement learning, we check whether reflection improves the performance of Q-learning on grid-based pathfinding environments. Specifically, we examine if the benefit of exploration and exploitation rate parameter adjustment outweighs the cost of reflection in returns and missed opportunities of movement. In this section, we present our results in terms of reflection frequency, (ALPHA) and (EPSILON) over time, episodic returns and success rate for each agent. The goal of this evaluation is to find which types of environments self-reflection leads to the greatest improvement in performance, where it is counterproductive, and how this trade-off changes across environments of varying complexity.

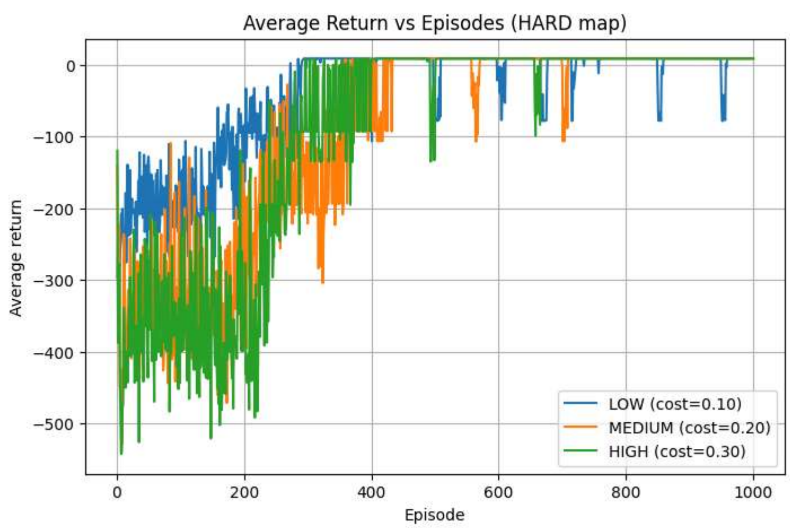
5.1 Performance by Difficulty Level

Episodic Returns



5. Evaluation





1. Easy Maps
   a. In easy environments, reflection provided no measurable benefit. The low-cost reflective agent had the most flections, while the medium-cost agent reflected moderately and the high-cost agent rarely chose to reflect. The baseline and high-cost agents obtained the highest success rates and the most favorable episodic returns, with the medium-cost agent performing similarly but not better than the baseline. The low-cost agent, despite frequent early reflection,

5. Evaluation

achieved the lowest episodic returns, indicating that the return cost of reflection and missed movement opportunity outweighed any learning advantage. All reflective agents learned over time to reduce the frequency of reflection, but this reduction did not translate into improvements over the baseline. These results suggest that in environments that are already easily solvable, reflection introduces unnecessary cost and does not provide strategic insight that improves learning outcomes.

2. Medium Maps
    a. In medium environments, the benefits of reflection started to show up, but behaved differently across cost levels. The low-cost reflective agent stopped reflecting earlier than the medium- and high-cost agents, while the high-cost agent continued reflecting throughout training, likely due to difficulty recovering from the more extreme adjustments to exploration and exploitation. Both low- and medium-cost agents achieved successful episodes, in contrast to the baseline agent, which did not solve the task within the allotted steps significantly. These results suggest that reflection can help to overcome challenges with minimal rewards, although excessive cost may prevent the agent's ability to escape suboptimal behavior once reflection is repeatedly punished.
3. Hard Maps
    a. Reflection demonstrated its greatest utility in high-difficulty environments. Both the low- and medium-cost agents achieved the highest success rates and episodic returns, substantially outperforming the baseline agent, which failed to complete the task in the given training horizon on average. The low-cost agent reflected frequently in the early episodes but gradually reduced reflection once a reliable path to the goal was discovered, leading to consistently higher returns. The medium-cost agent followed a similar pattern, though with fewer total reflections. These findings indicate that reflection can speed up learning in environments that are not trivial but rather complex.

5. Evaluation

4. Summary
   a. Across map difficulties, reflection demonstrates value only when the environment is sufficiently complex and requires more deliberate exploration.
   b. In easy environments, reflection offers no benefit; agents quickly learn that revising decisions is unnecessary, and performance decreases when reflection incurs avoidable cost.
   c. In medium and hard environments, reflection directly improves learning by enabling agents to increase or decrease their exploration and learning rate in order to find a solution path, leading to higher rewards and successful paths that the baseline almost never discovers.
   d. Reflection helps learning by speeding up correction of bad actions, reducing repeated mistakes, and enabling earlier convergence to useful behaviors.
   e. Adjusting (ALPHA) and (EPSILON) speeds up convergence rate to a solution by tuning the learning rate and exploration rate to encourage such solutions.

5. Evaluation

[DEPRECATED] Experimental Setup

We place all agents within 20 x 20 randomly generated pathfinding environments, each containing 100-200 randomly placed obstacles. The start and goal positions are placed such that there is a minimum of 15 steps between them. All agents have access to 4 movement actions (up, down, left, right), and reflective agents have an additional self-reflection action. Movements that place an object in an obstacle are not allowed, and will cause the agent to choose another action. A movement action has a reward of -0.1, while self-reflection has a larger cost depending on the agent.

We test four agents: a baseline Q-learning agent with no reflection, a low cost agent with a reflective reward of -0.15, a medium cost agent with a reflective reward of -0.25, and a high cost agent with a reflective cost of -0.35. Self-reflection changes the agent's learning process by adjusting both the learning rate (ALPHA SYMBOL) and the exploration rate (EPSILON SYMBOL). The reflection algorithm compares the return of the most recent steps with the steps preceding it. An improvement will result in a decrease of exploration rate and an increase of learning rate, while a decrease in returns results in an increase of exploration rate and a decrease of learning rate, as seen in the algorithm below.

(INSERT REFLECTION ALGORITHM)

To examine performance across a range of difficulties, we categorize environments into three difficulty levels, easy, medium, and hard. Each environment is first evaluated using the baseline agent for three independent trials of 300 episodes. Environments in which the baseline success rate exceeds 95% are classified as easy, those with success rates less than or equal to 25% are classified as hard, and all remaining environments are labeled medium. All reported results are averaged to minimize the variance of reinforcement learning.

[OL]6 Discussion

We have established a functioning single-agent environment in Gridworld where reflection is an additional action incurring a cost.

From the initial results, we can say that the environment and algorithm behave as expected. The agents learn to maximize reward while avoiding costly steps. However, because the reflection operator F(π) is not yet performing actual self-analysis, the results currently capture only the agent's initial learning behaviour before self-reflection is actually implemented rather than the intended cognitive trade-off.

Based on the results from the first version of our program, we can see that adding reflection as an explicit action is compatible with standard reinforcement-learning formulations and provides a flexible framework for extending to meta-learning experiments. However, these are only our early findings and interpretability is limited because reflection currently has no positive utility.

[OL] 7 Future Work

A key area of exploration would be to formulate and adopt a more robust model of reflection. In particular, we would allow agents to articulate and interpret aspects of their own policy in addition to their past actions. This formulation would encompass more of the widely-used definitions of reflection, leading to more general results.

This project could also be expanded to include multiple agents instead of just one. Each agent will have its own learning process and reflection steps, but they will share the same environment. Agents can either cooperate or complement each other while learning. This setting also raises the new possibility of shared reflection between agents by empowering agents to consider the past trajectories of other agents. By expanding our experiments to multi-agent settings, we would be able to better understand how reflection spreads between agents and how more opportunities for reflection affects overall learning speed.

Another goal for future work is allow agents to decide when to reflect and when to act. Agents would be able to choose whether to share their reflections or keep them private. This formulation could serve as a better model of the real-world behavior of robots or artificial intelligence systems.

[OL] 8 Conclusion

This project investigated whether adding an explicit reflect action to a standard reinforcement-learning agent can improve performance despite an immediate cost. We compared a baseline tabular Q-learning agent that can only take directional actions (up, down, left, right) to a reflective variant that can additionally choose reflect. When the agent reflects, the environment state remains unchanged, the agent receives an immediate penalty −c-c−c, and the agent adjusts its internal learning behavior by updating the exploration rate ϵ\epsilonϵ and learning rate α\alphaα using a recent-performance comparison.

Across experiments in GridWorld, we evaluated episodic return and how frequently the agent chose to reflect under multiple reflection-cost settings. The results indicate that reflection is not universally beneficial: its utility depends on whether the adaptive updates to ϵ\epsilonϵ and α\alphaα produce improvements large enough to outweigh the reflection cost. As the reflection cost increases, the agent has less incentive to choose the reflect action, so reflection becomes less frequent and any benefits must be large enough to offset the added penalty; when the cost is low, the penalty is smaller, making reflection more attractive and potentially beneficial if the α/ϵ adjustments improve subsequent decisions.

Two validation points are important for interpreting the findings. First, if the maps are categorized as easy, medium, and hard, these categories are verified empirically for the baseline agent to ensure they match actual difficulty. Second, to confirm that performance changes are truly caused by reflection's internal parameter updates, an ablation where reflect incurs the same cost but does not modify α\alphaα or ϵ\epsilonϵ would provide a clean control. Overall, this work supports the view that self-reflection is best understood as a cost-sensitive mechanism: it can be beneficial in regimes where adaptive learning is needed, but it can become neutral or detrimental when its cost outweighs its benefits.

8[OL] Shortcomings

The reflection mechanism is narrowly defined: reflection only increases the exploration rate after poor performance, a change that neither reliably improves learning nor meaningfully summarizes past behavior. Under this formulation, the agent has little reason to reflect, since the only modeled downside is the explicit reflection cost. Without any additional risks or long-term advantages tied to reflection, the agent cannot learn a compelling strategic use for the reflect action.

Furthermore, the use of tabular Q-learning and coordinate-based state representations limits both task complexity and the agent's capacity to generalize. Because states encode only positions rather than structural features of the environment, the agent cannot connect successful reflection in one region of the map to similar situations elsewhere. Reflection, therefore, becomes beneficial only after the optimal path is already discovered—at which point the agent no longer needs to reflect, making the mechanism effectively redundant in practice.

The main drawbacks of this experiment can be divided into three categories: first, the implementation-specific issues of the reflection mechanism; second, the shortcomings arising from the design and selection of the reinforcement learning algorithm; and third, the issues related to the experimental environment.

From the perspective of the implementation of the reflection mechanism, there is a lack of strong support for using increasing the exploration rate as a reflection behavior when the algorithm performs poorly; that is, increasing the exploration rate must necessarily bring good results. Specifically, if it cannot be guaranteed that increasing the exploration rate will bring good results, then when the agent chooses to reflect, it may not be able to obtain rewards in the short or long term. In that case, the agent has no reason to choose the behavior of reflection, and studying whether the agent will choose to reflect based on long-term considerations becomes meaningless. In addition, in a narrow sense, reflection should include summarizing the past few steps, but the level of exploration rate may not necessarily have a decisive impact on past performance. Therefore, adjusting the exploration rate cannot be well regarded as a strategy modification based on past performance.

There are also some issues with the implementation and algorithm selection of reinforcement learning. Using the standard Q-Learning algorithm, to some extent, limits the capabilities of the baseline agent, preventing overly complex experimental settings that could hinder the baseline agent from achieving a basic success rate. However, the reflection mechanism, as a behavior that in principle has long-term benefits, might be more meaningful in more complex tasks. Furthermore, using grid coordinates instead of features as the state in the experiment reduces the likelihood of the agent choosing to reflect. To elaborate: if we use the map features of the current state as the state input, and if the agent gains benefits through reflection in a complex terrain, then it is more likely to choose reflective behavior the next time it encounters similar terrain. However, when states are simply recorded as coordinates, the agent can only consider the impact of previous reflections and choose whether to use them when it returns to those coordinates. The condition for an agent to choose to reflect is that it can recognize from previous records that reflecting at this position is most likely to lead to the correct path. But for basic Q-learning, to reach such a judgment, the correct path must have already been traversed. Therefore, if the Q-table already records the correct path, the agent doesn't need to choose to reflect again, especially when reflection only improves the exploration rate rather than updating to a better search strategy.

From the perspective of experimental environment setup, this experiment uses the number of obstacles as the sole criterion for judging the complexity of the experimental environment. However, in actual implementation, there may be a possibility that too many obstacles will result in a particularly clear and direct path between the starting point and the destination. Specifically, the increase in obstacles reduces the area that the agent can explore, and the entire reachable area may be easier to search, making pathfinding easier.

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